Optimal Constellation Mapping Input: Specification of the BLOSUM62-Derived Sign Similarity Matrix ($\mathbf{S}$)

I. Foundational Principles of Biological Constellation Mapping

The Optimal Constellation Mapping (OCM) Protocol is an advanced optimization framework designed to embed the functional landscape of the standard genetic code onto a geometric structure, specifically the 6-dimensional hypercube, or Hexeract ($\mathbb{F}_2^6$). The fundamental objective is to identify a non-lexicographical binary encoding ($\mathbf{M}_{\text{opt}}$) for the 64 codons such that the resulting geometric constellation optimally mirrors biological and evolutionary similarity. This process requires a precisely defined measure of biological affinity, formalized as the Sign Similarity Matrix ($\mathbf{S}$).

1.1. The Constraint Space: Hexeract Geometry and Hamming Distance

The genetic code consists of $4^3 = 64$ unique codons. The minimum integer bit-space required to assign a unique address to each codon is six bits ($2^6 = 64$). This space defines the Hexeract, where each node represents a unique 6-bit binary encoding $\mathbf{M}_i$.
The metric used to quantify geometric proximity within the Hexeract is the Hamming distance, $H(\mathbf{M}_i, \mathbf{M}_j)$, which counts the number of differing bits between two 6-bit vectors $\mathbf{M}_i$ and $\mathbf{M}_j$. The OCM protocol mandates that the resulting topology $\mathbf{M}_{\text{opt}}$ ensures codons with high functional similarity are mapped to nodes separated by minimal Hamming distances (typically $H=1$ or $H=2$). Conversely, functionally dissimilar codons must be geometrically remote, exhibiting large Hamming distances. This objective establishes the foundation for the cost function $E(\mathbf{M})$, which links geometric separation H with biological cost derived from $\mathbf{S}$.

1.2. Robustness of the Genetic Code (RGC) and Cost Function Design

The Standard Genetic Code (SGC) is recognized for its profound robustness against point mutations. Single-nucleotide substitutions are statistically biased toward resulting in synonymous changes (coding for the same amino acid) or conservative non-synonymous changes (coding for an amino acid with similar physicochemical properties, such as replacing Leucine with Isoleucine). This intrinsic property, known as the Robustness of the Genetic Code (RGC), is the critical evolutionary feature the OCM must reproduce geometrically.
To effectively model RGC robustness, the cost function must utilize a similarity metric $\mathbf{S}$ that accurately reflects empirically observed evolutionary exchangeability. If a functional pair (R_i, R_j) possesses high biological similarity (high $\mathbf{S}_{ij}$), the cost function must severely penalize geometric separation (large $H(\mathbf{M}_i, \mathbf{M}_j)$). The selection of an evolutionary substitution matrix, such as BLOSUM, as the basis for $\mathbf{S}$, is therefore

fundamentally justified: it quantifies the functional compatibility required for survival across evolutionary time.
The OCM execution is conceptually equivalent to solving a weighted graph embedding problem. The system operates on two distinct graphs: the Biological Similarity Graph (G_B), where edges are weighted by the similarities $\mathbf{S}_{ij}$, and the Hexeract Graph (G_H), where edges are weighted by the geometric Hamming distance H. The optimization seeks a node mapping $\mathbf{M}$ that minimizes the product of these weights summed across all pairs. If the input matrix $\mathbf{S}$ correctly quantifies the necessary biological proximity or relationship, the optimal map $\mathbf{M}_{\text{opt}}$ will achieve a strong correlation between high biological affinity and minimal geometric distance. This process ensures that the resulting 6-cube constellation is not an arbitrary clustering but rather an accurate topological representation of observed evolutionary pathways and functional requirements.

1.3. Codon Representation and Indexing Scheme

For the 64×64 Sign Similarity Matrix ($\mathbf{S}$) to be processed consistently by the Octeract Cypher framework, a rigorous indexing convention must be established for the 64 codons (C_i). The standard genetic code (Code 1) is utilized, employing DNA bases (T, C, A, G) instead of RNA bases (U, C, A, G) for consistency with common sequence alignment formats. The standard lexicographical indexing scheme for $i \in \{1, 2, \ldots, 64\}$ is defined by the 5' $\rightarrow$ 3' direction, using the base order T, C, A, G for each position, with Base 3 varying fastest, followed by Base 2, and Base 1 slowest. This provides the canonical order necessary for matrix construction.
Table I defines the indices and corresponding translation products used to generate the $\mathbf{S}$ matrix.
Table I: Standard Genetic Code Mapping and Indexing (Partial List)

Index i	**Codon** C_i (DNA)	**Base 1**	**Base 2**	**Base 3**	**Residue** R_i (Full Name)	**One-Letter Code**
1	TTT	T	T	T	Phenylalanine	F
2	TTC	T	T	C	Phenylalanine	F
3	TTA	T	T	A	Leucine	L
4	TTG	T	T	G	Leucine	L
5	TCT	T	C	T	Serine	S
6	TCC	T	C	C	Serine	S
7	TCA	T	C	A	Serine	S
8	TCG	T	C	G	Serine	S
...	...	...	...	...	...	...
49	GTT	G	T	T	Valine	V
50	GTC	G	T	C	Valine	V
51	GTA	G	T	A	Valine	V
52	GTG	G	T	G	Valine	V
...	...	...	...	...	...	...
64	GGG	G	G	G	Glycine	G

II. Metric Specification: The Codon Similarity Score

(\mathbf{S})

2.1. Selection of the Primary Metric: BLOSUM62

The chosen primary metric for quantifying biological similarity and constructing the \mathbf{S} matrix is the **BLOSUM62** substitution matrix. This choice is dictated by the requirement that the metric reflects empirically observed functional interchangeability under evolutionary pressure. BLOSUM matrices, unlike PAM matrices which are extrapolated from closely related sequences, are derived directly from substitution frequencies found within "blocks" of local alignments from sequences exhibiting no more than 62% sequence identity. BLOSUM62 is the standard matrix used in protein BLAST searches and is optimized for identifying homologous proteins over intermediate evolutionary distances. Its derivation from observed substitutions in divergent sequences provides a robust log-odds measure of residue exchangeability, directly translating biological constraint into a numerical score.

2.2. The BLOSUM62 Core Matrix (\mathbf{A}_{21 \times 21})

The foundation of the \mathbf{S} matrix is the 21 \times 21 core amino acid substitution matrix, designated \mathbf{A}. The entries \mathbf{A}_{xy} represent the log-odds score for aligning residue x with residue y based on the BLOSUM62 model.
The numerical range of \mathbf{A} extends from +11 to -4. Positive scores signify conservative substitutions that occur frequently in blocks of homologous proteins, suggesting that the functional and physicochemical properties are highly compatible (e.g., Leucine/Isoleucine exchange scores +2). The highest score of +11 is assigned to Tryptophan aligned with Tryptophan (W:W), highlighting its extreme evolutionary conservation. Negative scores, conversely, classify non-conservative, functionally disruptive substitutions (e.g., Asparagine/Tryptophan scores -4). These scores are scaled in half-bit units, providing a measure of how much more likely a substitution is in related sequences than by chance.
The matrix \mathbf{A} must include entries for all 20 standard amino acids and the essential translation termination signal, the STOP codon (*).
Table II: BLOSUM62 21 \times 21 Core Substitution Matrix (\mathbf{A})

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V	*
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0	-4
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3	-4
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3	-4
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3	-4
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1	-4
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2	-4
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2	-4
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3	-4
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3	-4
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3	-4
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1	-4
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2	-4
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1	-4
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1	-4
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2	-4
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2	-4
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0	-4

W | -3 | -3 | -4 | -4 | -2 | -2 | -3 | -2 | -2 | -3 | -2 | -3 | -1 | 1 | -4 | -3 | -2 | 11 | 2 | -3 | -4 | | **Y** | -2 | -2 | -2 | -3 | -2 | -1 | -2 | -3 | 2 | -1 | -1 | -2 | -1 | 3 | -3 | -2 | -2 | 2 | 7 | -1 | -4 | | **V** | 0 | -3 | -3 | -3 | -1 | -2 | -2 | -3 | -3 | 3 | 1 | -2 | 1 | -1 | -2 | -2 | 0 | -3 | -1 | 4 | -4 | | ***** | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 | 1 |
(Data derived from the 21 \times 21 portion of the BLOSUM62 matrix)

2.3. Nuanced Handling of the STOP Codon Functional Distance

The handling of the STOP codon (specifically TGA, TAA, and TAG) within the similarity matrix is critical, as it defines the boundary between functional coding space and termination signals. A substitution that converts a coding amino acid into a STOP codon, or vice versa (read-through mutation), represents a catastrophic failure of the protein sequence.
Standard alignment algorithms capture this functional distance using a severe penalty. In the BLOSUM62 core matrix \mathbf{A}, the substitution score between any amino acid X and the STOP codon (*) is fixed at \mathbf{A}_{X, *} = -4. This value is the maximum negative penalty assigned to any substitution pairing in the matrix (excluding gap penalties which are contextual), confirming its status as a functionally disruptive event. This universal penalty guarantees that, in the resultant OCM, all coding codon clusters are geometrically separated from the three non-sense codons.
An important subtlety lies in the self-similarity score for the STOP codon: \mathbf{A}_{*, *} = +1. If this score were as high as other highly conserved amino acids (e.g., W:W = +11), the optimization would prioritize placing the three distinct STOP codons into an extremely tight geometric cluster, potentially consuming geometric edges better used for highly redundant synonymous amino acid groups (like Serine, which has six codons). The modest self-score of +1 ensures that while the three STOP codons retain geometric cohesion relative to the large X \leftrightarrow * penalty, they do not exert excessive geometric pressure on the optimization, allowing the system to use the limited number of low Hamming distance edges (H=1) primarily to map biological pathways between conserved amino acids.

III. Construction of the 64 \times 64 Codon Similarity Matrix (\mathbf{S})

3.1. Defining the Mapping Function M(C)

The construction of the 64 \times 64 matrix \mathbf{S} relies entirely on the translation function M(C). This function maps any input codon C_i to one of the 21 possible translation products R \in \{A, R, N, D, C, Q, E, G, H, I, L, K, M, F, P, S, T, W, Y, V, *\}. This mapping is derived from the Standard Genetic Code (Table I).

3.2. Rule Set for Codon-to-Codon Similarity \mathbf{S}_{ij}

The 64 \times 64 Sign Similarity Matrix \mathbf{S} quantifies the biological similarity between every possible pair of codons (C_i, C_j). Since biological function is defined by the translated product, \mathbf{S}_{ij} is defined solely by the evolutionary substitution score between the resulting residues, R_i and R_j.
Formal Definition: Let C_i and C_j be Codon i and Codon j. Let R_i = M(C_i) and R_j = M(C_j)

be their respective translated products. The similarity score $\mathbf{S}_{ij}$ is defined as:
This matrix contains 4,096 entries, of which 1,088 entries represent synonymous or conservative non-synonymous substitutions (positive $\mathbf{S}$ values), and 3,008 entries represent disruptive or non-sense substitutions (negative $\mathbf{S}$ values).

Analysis of Substitution Cases:

1. **Synonymous Codons ($R_i = R_j$):** If the two codons translate to the same residue (e.g., GGT and GGC both translate to Glycine, G), the similarity score $\mathbf{S}_{ij}$ is the self-identity score for that amino acid ($\mathbf{A}_{G, G} = +6$). These high positive scores generate maximum pressure on the OCM algorithm to place these codons into the tightest possible geometric clusters in the Hexeract (ideally H=1 or H=2).
2. **Non-Synonymous, Non-Sense Codons ($R_i \neq *$, $R_j \neq *$):** If the codons translate to two different active amino acids, the score $\mathbf{S}_{ij}$ reflects the functional difference between R_i and R_j. For instance, the substitution from Isoleucine (I) to Valine (V) is conservative, yielding $\mathbf{S}_{IV} = +3$. Conversely, a disruptive change, such as Glycine (G) to Glutamate (E), yields $\mathbf{S}_{GE} = -2$. The OCM protocol leverages this range: conservative changes demand minimal geometric distance, ensuring that physicochemical similarity is geometrically respected, while disruptive changes tolerate maximal geometric distance, facilitating separation of functionally distant groups.
3. **Non-Sense Codons (STOP Codons):**
    - **AA $\leftrightarrow$ STOP:** If one codon codes for an amino acid and the other is a STOP codon, $\mathbf{S}_{ij}$ is universally -4. This severe, constant penalty ensures the three STOP codons are driven to the geometric periphery, maintaining maximum Hamming distance from the vast clusters of functional coding codons.
    - **STOP $\leftrightarrow$ STOP:** If both codons are STOP codons, $\mathbf{S}_{ij} = +1$. This positive score is sufficient to ensure the three non-sense codons are geometrically proximate to each other.

The definition of $\mathbf{S}_{ij}$ based purely on the translated product ($\mathbf{A}_{R_i, R_j}$) intentionally decouples the similarity metric from the nucleotide-level Hamming distance ($H_N(C_i, C_j)$). This decoupling is essential to the OCM's effectiveness, as the optimization process is tasked with *discovering* the underlying binary code $\mathbf{M}_{\text{opt}}$ that minimizes cost. If $\mathbf{S}$ were pre-biased by H_N, the resulting geometric map would merely reflect the existing, arbitrary binary assignment of T, C, A, G, defeating the goal of finding an optimal functional embedding. By focusing solely on the functional output ($\mathbf{A}$), the OCM is forced to map the evolutionary robustness (high $\mathbf{S}$ for single-base change codons) onto the minimal geometric distances (H=1), thereby validating the hypothesis that the functional constraints of the SGC can be accurately represented topologically in the Hexeract.

3.3. Algorithmic Implementation of $\mathbf{S}$ Generation

The generation of $\mathbf{S}$ requires the implementation of the translation mapping function M(C) and the $\mathbf{A}$ lookup table within the Octeract system.
The steps are:

1. Input the 21×21 BLOSUM62 matrix $\mathbf{A}$.
2. Input the $64 \rightarrow 21$ translation table M(C).
3. Iteratively calculate the 64×64 matrix $\mathbf{S}$ using the rule $\mathbf{S}_{ij} = \mathbf{A}_{M(C_i), M(C_j)}$.

Table III provides examples of how the similarity scores $\mathbf{S}_{ij}$ are derived for various

substitution types using the core $\mathbf{A}$ matrix values presented in Table II.
Table III: Sample Codon Pair Similarity Derivation

Codon C_i	**Codon** C_j	**Residue** R_i	**Residue** R_j	$\mathbf{A}_{R_i, R_j}$ (BLOSUM62)	$\mathbf{S}_{ij}$ (Similarity)	**Substitution Type**	$\mathbf{S}^*_{ij}$ (Cost)
GGT (Gly)	GGC (Gly)	G	G	+6	+6	Synonymous	5
ATT (Ile)	GTT (Val)	I	V	+3	+3	Conservative Non-Synonymous	8
TCT (Ser)	TGT (Cys)	S	C	-1	-1	Moderately Disruptive Non-Synonymous	12
GGT (Gly)	GAA (Glu)	G	E	-2	-2	Disruptive Non-Synonymous	13
GGT (Gly)	TGA (STOP)	G	*	-4	-4	Non-Sense (Catastrophic)	15
TAG (STOP)	TAA (STOP)	*	*	+1	+1	Non-Sense Cohesion	10

IV. Optimization and Cost Function Integration for OCM

The Sign Similarity Matrix $\mathbf{S}$ provides the raw measure of biological affinity. For the Optimal Constellation Mapping (OCM) protocol, which utilizes the Simulated Annealing heuristic, this similarity must be converted into a cost or energy function $E(\mathbf{M})$ that is minimized during the optimization process.

4.1. Formal Definition of the OCM Cost Function $E(\mathbf{M})$

The OCM cost function calculates the total energy of a given codon encoding $\mathbf{M}$ by summing the products of the biological cost ($\mathbf{S}^*_{ij}$) and the geometric distance ($H(\mathbf{M}_i, \mathbf{M}_j)$) across all codon pairs:
The optimal mapping $\mathbf{M}_{\text{opt}}$ is the encoding that yields the minimum possible $E(\mathbf{M})$.

4.2. Transformation of Similarity to Cost ($\mathbf{S} \rightarrow \mathbf{S}^*$)

Since $\mathbf{S}$ represents similarity (where high values are desirable), it must be inverted to create the cost matrix $\mathbf{S}^*$ (where high values are undesirable, representing high penalty).

The range of the BLOSUM62 scores \mathbf{S} is R = [-4, +11]. The maximum possible similarity, S_{\text{max}}, is +11. The linear transformation ensures that the highest similarity is mapped to the lowest cost, and the lowest similarity is mapped to the highest cost.

Transformation Rule:

\mathbf{S}^**{ij} = 11 - \mathbf{S}*{ij}$$

Cost Matrix Range:

1. **Maximum Similarity (\mathbf{S}_{ij} = +11, e.g., W:W):** Results in minimum cost: \mathbf{S}^*_{ij} = 11 - 11 = 0. (These codons should ideally occupy the same space, H=0, if i=j, or be highly proximate if i \neq j and \mathbf{S}_{ij} is high, such as synonymous pairs).
2. **Minimum Similarity (\mathbf{S}_{ij} = -4, e.g., A \leftrightarrow *):** Results in maximum cost: \mathbf{S}^*_{ij} = 11 - (-4) = 15. (These catastrophic substitution pairs incur the greatest penalty and must be separated by the maximum possible Hamming distance, H=6, to minimize energy contribution).

Final Operational Cost Function: The OCM Simulated Annealing algorithm will minimize the total energy defined by:

The magnitude of the cost matrix \mathbf{S}^* (ranging from 0 to 15) provides the strong weighting factor needed to guide the optimization process. For instance, if two codons that translate to Glycine and Glutamate (\mathbf{S}_{GE} = -2, \mathbf{S}^*_{GE} = 13) are placed at H=1, they contribute 13 energy units. If two synonymous codons (Gly:Gly, \mathbf{S}^*=5) are separated by H=6, they contribute 30 energy units. This high penalty for separating functionally related groups is the primary driver ensuring that functional clusters are geometrically condensed. The overwhelming maximum cost contribution of 90 units (derived from \mathbf{S}^*=15 separated by H=6) acts as the dominant force compelling the three non-sense codons to the most distant geometric corners of the Hexeract relative to the functional coding space.

4.3. Scaling and Normalization for Simulated Annealing

The \mathbf{S}^* scores, ranging from 0 to 15, are proportional to the fundamental log-odds scale of BLOSUM62 (half-bit units). Given that the maximum potential individual cost contribution is 90 (15 \times 6), and the total number of pairs is 4,096, the total energy E(\mathbf{M}) can reach values approaching 370,000 (though a realistic, sub-optimal encoding will be significantly lower).

For computational stability within the Simulated Annealing framework, the integer scale (0-15) of \mathbf{S}^* is maintained, as it preserves the evolutionary significance of the BLOSUM log-odds ratios. Instead of explicit normalization, the system relies on tailoring the Simulated Annealing temperature schedule to the energy scale. An initial temperature (T_0) sufficient to accept unfavorable moves, estimated to be in the range of 500 to 1000, must be employed. A slow geometric cooling schedule (e.g., T_{k+1} = \alpha T_k with \alpha \approx 0.999) is critical to allow the complex system to thoroughly explore the 2^{64} phase space and avoid premature convergence in local energy minima.

V. Conclusion and Mandatory Input Summary

The input required for the Optimal Constellation Mapping (OCM) Simulated Annealing protocol is the 64 \times 64 Sign Similarity Matrix (\mathbf{S}) defined by the functional relatedness of

the translated amino acids.
The metric specification ensures that the geometric optimization is driven by rigorous, empirically derived evolutionary constraints, thereby guaranteeing that the resulting Hexeract constellation $\mathbf{M}_{\text{opt}}$ reflects the intrinsic robustness and functional organization of the Standard Genetic Code.

5.1. Mandatory Input Summary for OCM Protocol

The following input structures are required for the Octeract Cypher framework to execute the Simulated Annealing algorithm (Section 2.2 of the framework) and find the Optimal 6-Bit Mapping ($\mathbf{M}_{\text{opt}}$):

Input Component	Specification	Value/Definition
Metric Specification	Amino Acid Substitution Cost	BLOSUM62 (Evolutionary Log-Odds Matrix)
Core Data Matrix ($\mathbf{A}$)	21 $\times$ 21 BLOSUM62 Score Matrix	Provided in Table II
Codon Translation Function ($M(C)$)	Standard Genetic Code (Code 1, DNA)	Defined in Table I (64 inputs $\rightarrow$ 21 outputs)
Sign Similarity Matrix ($\mathbf{S}$)	64 $\times$ 64 Matrix	$\mathbf{S}_{ij} = \mathbf{A}_{M(C_i), M(C_j)}$
Cost Matrix Transformation ($\mathbf{S}^*$)	Linear Inversion of Similarity to Cost	$\mathbf{S}^*_{ij} = 11 - \mathbf{S}_{ij}$ (Range 0 to 15)
OCM Cost Function ($E(\mathbf{M})$)	Target Function to be Minimized	$E(\mathbf{M}) = \sum_{i} \sum_{j} (11 - \mathbf{S}_{ij}) \cdot H(\mathbf{M}_i, \mathbf{M}_j)$

Upon receiving this formalized input, the Simulated Annealing process will proceed to identify the encoding $\mathbf{M}_{\text{opt}}$ that minimizes total geometric distance for evolutionarily favorable codon pairs while maximizing distance for functionally disruptive or non-sense pairings. The successful convergence of this optimization will provide a geometrically optimal map where biological pathways (single-step mutations resulting in conservative changes) appear naturally along the shortest geometric edges (H=1 or H=2) of the Hexeract.

Works cited

1. Substitution matrix - Wikipedia, https://en.wikipedia.org/wiki/Substitution_matrix 2. BLOSUM62 Substitution Matrix - LabXchange, https://www.labxchange.org/library/items/lb:LabXchange:24d0ec21:lx_image:1 3. DNA and RNA codon tables - Wikipedia, https://en.wikipedia.org/wiki/DNA_and_RNA_codon_tables 4. Genetic Code Translation Tables - NCBI, https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi 5. Scoring matrices - QIAGEN Bioinformatics Manuals, https://resources.qiagenbioinformatics.com/manuals/clcgenomicsworkbench/2203/index.php?manual=BE_Scoring_matrices.html 6. BLOSUM - Wikipedia, https://en.wikipedia.org/wiki/BLOSUM 7. Selecting the Right Similarity-Scoring Matrix - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC3848038/ 8. /c/data/BLOSUM62 - NCBI, https://www.ncbi.nlm.nih.gov/IEB/ToolBox/C_DOC/lxr/source/data/BLOSUM62 9. Sequence Alignment - Handbook of Discrete and Combinatorial Mathematics - NCBI - NIH,

https://www.ncbi.nlm.nih.gov/books/NBK464187/